

Rugby Intelligent Camera

*Note: Sub-titles are not captured in Xplore and should not be used

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Abstract—Camera-based tracking systems are increasingly used in sports to enhance performance analysis and broadcast production. However, existing solutions are typically expensive and complex, making them inaccessible for amateur clubs with limited resources. This project aims to develop a cost-effective and functional camera tracking solution to, in real time, broadcast and capture the statistics of an amateur rugby match. Our proposition relies on a multi-camera setup which is controlled by a trained computer vision model, YOLO (You Only Look Once). The conducted tests show an accuracy of 80

Index Terms—computer vision, rugby, cost-effective, camera tracking, YOLO

I. INTRODUCTION

This document is a model and instructions for L^AT_EX. Please observe the conference page limits.

II. TECHNICAL CHOISES (EXPLAINING HOW WE APPROCHED THE PROBLEM)

A. Maintaining the Integrity of the Specifications

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III. OUR WORK

We can change the order of the project aswell

A. YOLO Model

To track the focus of a rugby match, we agreed that the most important thing we should track is the ball, but in rugby the ball is small and always hidden between the players or very close to their chest, causing a big problem for the model's identification of the ball. In order to solve this problem, we added the class player; this implies that the players will also be identified by the model.

In rugby, the teams usually place themselves in lines of attack and defense, so, normally the point of interest will take

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place where the greatest number of players are. With both classes, we can already create a working tracking model.

The world of computer vision developers is growing and one of the biggest communities is the Roboflow Universe, which we can find a great number of models and datasets. The project Rugby Detection Computer Vision Model [8] was the closest to our project's needs, and the pictures used in this dataset are similar to the images we captured in an INSA AS match.

After testing the model from the project [8] in some images from the captured match (Fig. 1 to Fig. 3), positive results were achieved: the model responds well to the images, identifying a big number of player and the ball. Some problems were: confusing the arbitrator with a player, not identifying all the players, and sometimes not seeing the ball.

Unfortunately, the model [8] isn't accessible to download, so it wouldn't be possible to use in a real time application, because for every frame the Roboflow repository would have to be accessed on the internet, increasing the cost and delay of the image processing.

To solve the presented problem, the project's dataset [8] was cloned and a new AI model was trained, hoping to achieve the same results as the previous. Four models were created, two with yolov8n, one with yolov8s and the last with yolov8m; they were all tested in an image not from the dataset [??? Show images of the test???].

These models aren't as good as the previous one, but they are functional for video analyses and can be used for real-time applications.

IMAGES COMPARISON

To validate our models, we tested it in a video of an INSA AS match, but the power of the available GPU wasn't enough to analyze a nineteen second video (with 1141 frames). So, we couldn't validate our models in a hole video but, analyzing in a smaller video, we can see some problems of players and ball identification, showing how the project would look like in a real-time solution.



Fig. 1. Test image 1, used model from Roboflow [8].



Fig. 2. Test image 2, used model from Roboflow [8].

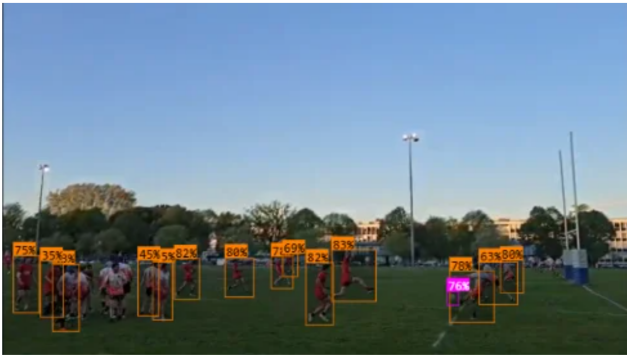


Fig. 3. Test image 3, used model from Roboflow [8].

Regarding the YOLO model we can conclude that this solution works for identification of our needs to choose the best camera but, trying to find a low-cost solution for the amateur clubs we arrived at a lack of computing power that probably will increase the cost of this approach.

To improve this project, we plan on using another YOLO model [9] to see the statistics of the game as well, identifying tackle, penalty kick, throw-in, scrum, and other similar rugby situations that our other model can't identify. Another aimed improvement is to create a system [9] so the model could be able to after identifying the players, separate them between

team A and team B. The system consists of a Sigmoid loss language image pre-trained [9] (extract color information from the player's shirt), associated with an UMAP [9] (compact the previous information) and a Kmeans [9] (sort both teams). These improvements would highly increase the quality of the performance analysis of the teams.

B. Video Positioning teams part

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C. Video Transmission teams part

Number equations consecutively. To make your

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Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not "Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

D. Camera score algorithm

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead

E. IHM

- The word "data" is plural, not singular.

An excellent style manual for science writers is [7].

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a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 4”, even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
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^aSample of a Table footnote.



Fig. 4. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

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